KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 7 REPORT – DIMENSIONALITY REDUCTION

Feature Selection, Principal Component Analysis and Variance Visualization

Youth Smoking and Drug Abuse Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

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1. Introduction

This report presents the Week 7 work of the project “Youth Smoking and Drug Abuse Analysis”. The focus of this week is Dimensionality Reduction. It covers feature selection, variance thresholding, correlation-based feature selection, multicollinearity analysis using Variance Inflation Factor (VIF), Principal Component Analysis (PCA), explained variance and PCA visualizations.

The analysis was performed using the cleaned dataset. A separate copy was created, and the seven personality features were selected for dimensionality reduction. The selected data contains 1,885 records and 7 features.

2. Objective

To identify useful personality features, examine low variance and relationships among features, and reduce the seven-dimensional feature space using PCA while retaining at least 95% of the total variance.

3. Activities Performed

- Created a separate copy of the cleaned dataset and selected seven personality features.
- Applied Variance Thresholding with a threshold of 0.01.
- Performed correlation-based feature selection using a threshold of 0.80.
- Checked multicollinearity using Variance Inflation Factor (VIF).
- Standardized the selected features and applied Principal Component Analysis.
- Calculated individual and cumulative explained variance.
- Visualized feature variance, correlation, explained variance and the first two principal components.

4. Approach and Data Preparation

4.1 Selecting Personality Features

A separate dataset was created using df_dr = df_clean.copy(). The following seven personality features were selected: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS. The selected feature matrix contained 1,885 rows and 7 columns.

Dataset Detail	Result
Number of records	1,885
Selected features	7
Feature names	Nscore, Escore, Oscore, Ascore, Cscore, Impulsive, SS
Initial feature matrix shape	(1885, 7)

Table 1: Selected personality features

5. Feature Selection

5.1 Variance Thresholding

Variance Thresholding removes features that have very low variance. Such features contain almost constant values and may provide limited information. A threshold of 0.01 was used in the notebook. The variance of every selected personality feature was greater than this threshold.

Feature	Variance	Decision
Nscore	0.942604	Retained
Escore	0.994903	Retained
Oscore	0.992471	Retained
Ascore	0.994888	Retained
Cscore	0.995051	Retained
Impulsive	0.910945	Retained
SS	0.928720	Retained

Table 2: Variance values and feature selection result

All seven features were retained. No feature was removed because none had variance below or equal to the selected low-variance threshold.

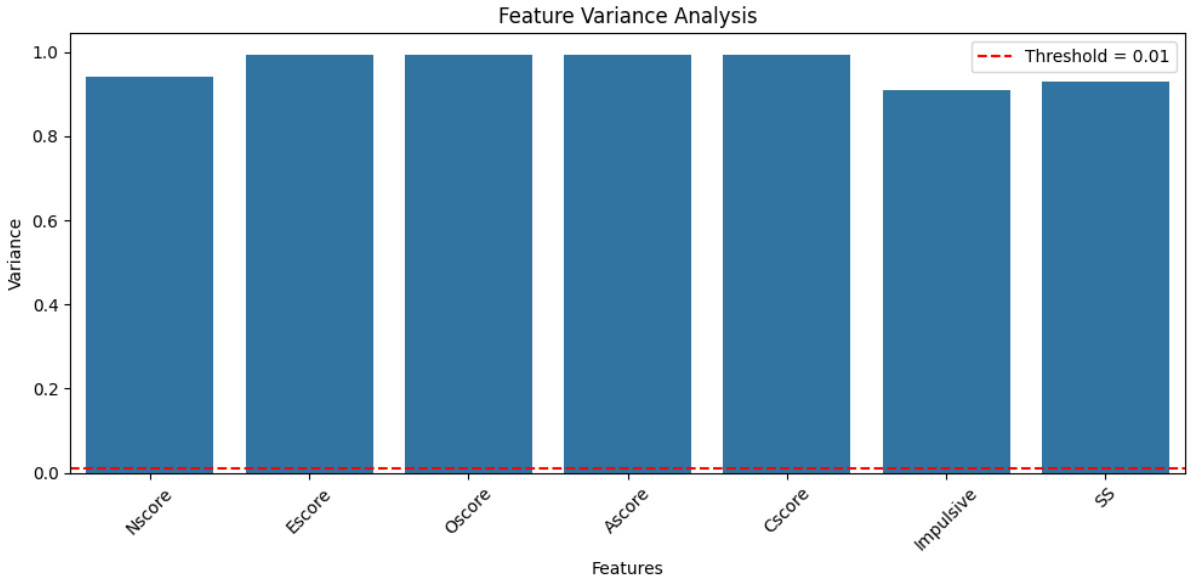


Figure 1: Feature variance analysis with the 0.01 threshold

5.2 Correlation-Based Feature Selection

Correlation-based feature selection checks whether two numerical features have a strong relationship. Highly correlated features may carry overlapping information. The notebook used an absolute correlation threshold of 0.80 to identify highly correlated pairs.

The correlation heatmap shows the relationships among the seven personality features. The highest observed correlation is between Impulsive and SS (0.623120), which is below 0.80. Therefore, no highly correlated feature pair was identified and no feature was removed by this check.

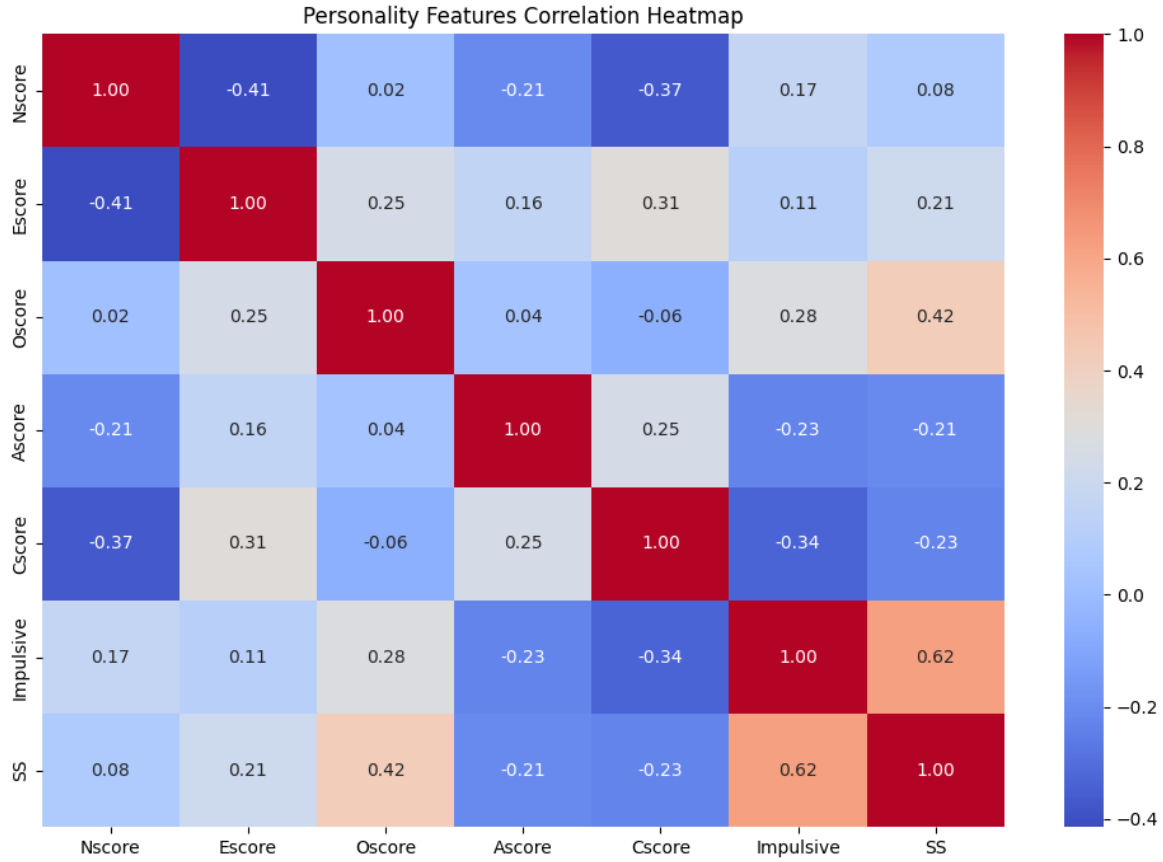


Figure 2: Correlation heatmap of the seven personality features

Feature Pair	Correlation	Observation
Impulsive – SS	0.623120	Highest positive correlation
Oscore – SS	0.421709	Moderate positive correlation
Nscore – Escore	-0.414434	Moderate negative correlation
Nscore – Cscore	-0.369426	Moderate negative correlation

Table 3: Selected correlation values

5.3 Multicollinearity Analysis using VIF

Multicollinearity occurs when independent features are strongly related to one another. Variance Inflation Factor (VIF) was calculated after standardizing the selected features. In this notebook, a VIF value greater than 5 was treated as a warning sign.

Feature	VIF	Result
Nscore	1.354701	Below 5
Escore	1.471786	Below 5

Oscore	1.280984	Below 5
Ascore	1.150772	Below 5
Cscore	1.389940	Below 5
Impulsive	1.790268	Below 5
SS	1.914818	Below 5

Table 4: Variance Inflation Factor values

All VIF values are below 5. The highest VIF is 1.914818 for SS. Hence, the notebook did not identify a feature that required removal due to multicollinearity.

6. Principal Component Analysis (PCA)

Principal Component Analysis converts the original features into new uncorrelated variables called principal components. Before applying PCA, the seven original personality features were standardized using StandardScaler. PCA was then applied to the standardized data.

The full PCA initially generated seven principal components from the original seven features. The components were then examined using their explained variance ratios to decide how many were needed to retain at least 95% of the information.

7. Explained Variance Analysis

Explained variance ratio indicates the proportion of total variance represented by each principal component. Cumulative explained variance adds these proportions one by one and shows the total variance retained as more components are included.

Principal Component	Explained Variance	Cumulative Variance
PC1	31.9857%	31.9857%
PC2	25.7709%	57.7566%
PC3	12.6940%	70.4506%
PC4	9.6546%	80.1052%
PC5	8.1591%	88.2643%
PC6	6.8041%	95.0684%
PC7	4.9316%	100.0000%

Table 5: Explained and cumulative variance of principal components

The first component explains about 31.99% of the variance. The first two components together explain about 57.76%. The cumulative variance reaches approximately 95.07% at PC6. Therefore, six principal components are required to retain at least 95% of the variance.

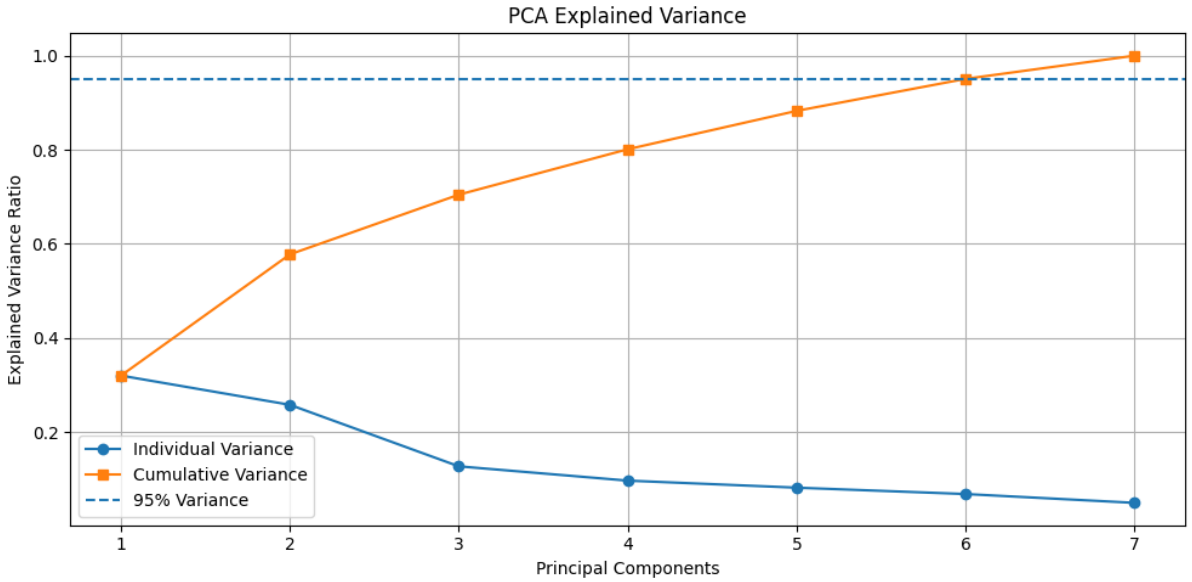


Figure 3: PCA explained variance – individual and cumulative variance

8. Dimensionality Reduction

Based on the cumulative explained variance, PCA was applied again with six components. The original feature matrix had 1,885 rows and 7 columns. The reduced dataset (df_pca) contains the same 1,885 records represented by 6 principal components.

Dataset	Rows	Columns
Original personality feature matrix	1,885	7
PCA reduced dataset	1,885	6

Table 6: Dataset shape before and after PCA

The dimensionality was reduced from seven features to six principal components while retaining approximately 95.07% of the total variance. This reduces the number of dimensions by one while preserving most of the information measured by variance.

9. Two-Dimensional PCA Visualization

The first two principal components (PC1 and PC2) were plotted in a scatter plot. Gender labels were added to distinguish observations in the visualization. This plot provides a two-dimensional view of the observations after PCA.

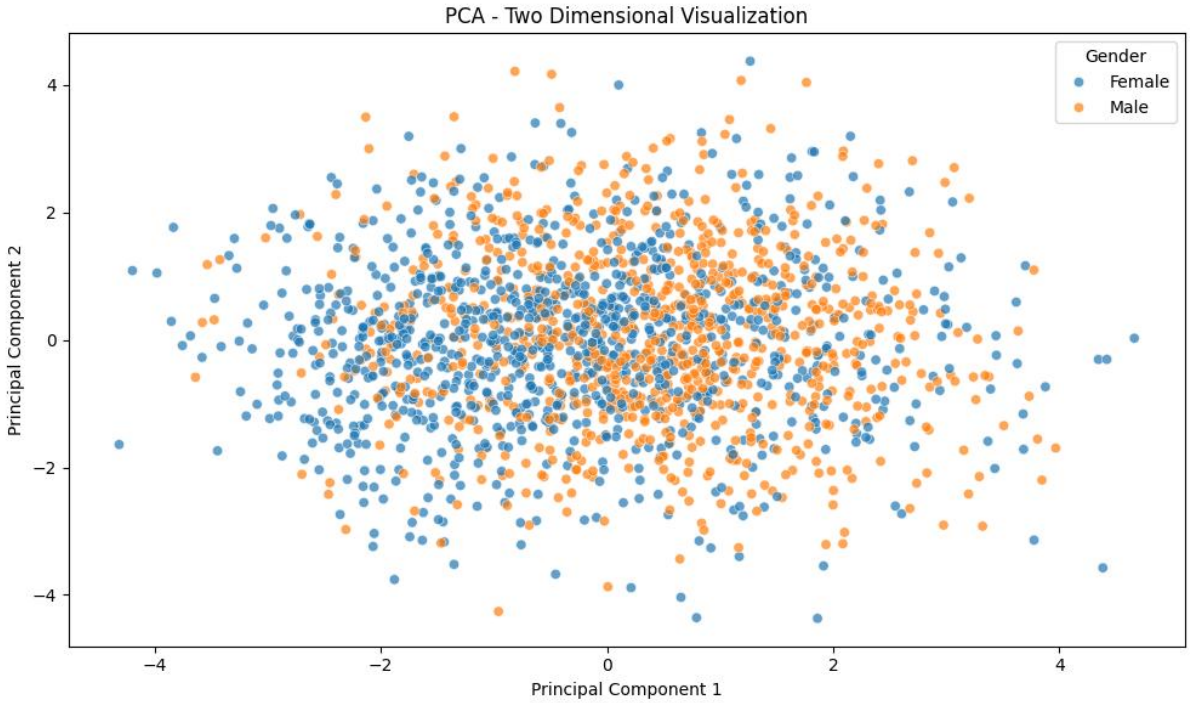


Figure 4: PCA two-dimensional visualization using PC1 and PC2, coloured by Gender

The scatter plot shows how the observations are distributed along the first two principal components. The first two components explain approximately 57.76% of the total variance together. The plot is useful for visual inspection, while the six-component dataset is used to retain at least 95% of the variance.

10. Key Findings

- The selected personality feature matrix contained 1,885 records and 7 features.
- Variance Thresholding with a threshold of 0.01 retained all seven features; none were removed.
- No feature pair exceeded the absolute correlation threshold of 0.80.
- All VIF values were below 5, so no high multicollinearity warning was found.
- PCA generated seven components from the standardized original features.
- The first six principal components retained 95.07% of the total variance.
- The reduced dataset contained 1,885 rows and 6 principal component columns.

11. Conclusion

Dimensionality Reduction was performed on the seven personality features of the Youth Smoking and Drug Abuse dataset. Feature selection checks using variance thresholding, correlation analysis and VIF showed that all seven original features could be retained. PCA was then applied to standardized features. The explained variance analysis showed that six principal components retain approximately 95.07% of the total variance. The PCA scatter plot of PC1 and PC2 provided a two-dimensional visualization of the observations. Overall, the notebook reduced the feature space from seven dimensions to six while preserving most of the information.